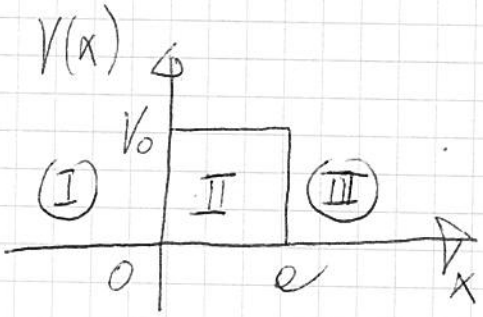

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{(dx)^2} = E\psi$$ NELLE REGIONI I E III $[x<0$ E $x>a]$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{(dx)^2} = (E-V_0)\psi$$ NELLA REGIONE II $[0<x<a]$

$$\psi(x) = \begin{cases} Ae^{ik_1x} + Be^{-ik_1x} & x<0 \quad k_1 = \sqrt{2mE}/\hbar \\ Fe^{-k_2x} + Ge^{k_2x} & a>x>0 \quad k_2 = \frac{\sqrt{2m(V_0-E)}}{\hbar} \\ Ce^{ik_1x} + De^{-ik_1x} & x>a \end{cases}$$

NOTA BENE, SIAMO NEL CASO $E<V_0$

INOLTRE $D=0$ POICHÉ L'ONDA INCIDE DA SINISTRA

QUINDI NELLA REGIONE III NON HO ONDA RIFLESSA

$$\psi(x=0^+) = \psi(x=0^-) \Rightarrow F+G = A+B \quad (1)$$

$$\psi(x=a^+) = \psi(x=a^-) \Rightarrow Ce^{ik_1a} = Fe^{-k_2a} + Ge^{k_2a} \quad (2)$$

$$\frac{d\psi}{dx}(x=0^+) = \frac{d\psi}{dx}(x=0^-) \Rightarrow ik_1A - ik_1B = -k_2F + k_2G \quad (3)$$

$$\frac{d\psi}{dx}(x=a^+) = \frac{d\psi}{dx}(x=a^-) \Rightarrow ik_1Ce^{ik_1a} = -k_2Fe^{-k_2a} + k_2Ge^{k_2a} \quad (4)$$

DOBBIAMO TROVARE $T = |C|^2/|A|^2$

(1) $A+B=F+G$

(3) $\frac{ik_1}{k_2}A-\frac{ik_1}{k_2}B=F-G$

(1)+(3) $A\left(1+\frac{ik_1}{k_2}\right)+B\left(1-\frac{ik_1}{k_2}\right)=2F$

(1)−(3) $A\left(1-\frac{ik_1}{k_2}\right)+B\left(1+\frac{ik_1}{k_2}\right)=2G$

(2) $Ce^{ik_1a}=Fe^{-k_2a}+Ge^{k_2a}$

(4) $i\frac{k_1}{k_2}Ce^{ik_1a}=-Fe^{-k_2a}+Ge^{k_2a}$

(2)+(4) $\left(1+\frac{ik_1}{k_2}\right)Ce^{ik_1a}=2Ge^{k_2a}$

(2)−(4) $\left(1-\frac{ik_1}{k_2}\right)Ce^{ik_1a}=2Fe^{-k_2a}$

$$A\left(1+\frac{ik_1}{k_2}\right)+B\left(1-\frac{ik_1}{k_2}\right)=\left(1-\frac{ik_1}{k_2}\right)Ce^{ik_1a}e^{k_2a}$$

$$A\left(1-\frac{ik_1}{k_2}\right)+B\left(1+\frac{ik_1}{k_2}\right)=\left(1+\frac{ik_1}{k_2}\right)Ce^{ik_1a}e^{-k_2a}$$

$$A\left(1+\frac{ik_1}{k_2}\right)\left(1-\frac{ik_1}{k_2}\right)^{-1}+B=Ce^{ik_1a}e^{k_2a}$$

$$A\left(1-\frac{ik_1}{k_2}\right)\left(1+\frac{ik_1}{k_2}\right)^{-1}+B=Ce^{ik_1a}e^{-k_2a}$$

$$A\left(\frac{k_2+ik_1}{k_2}\right)\left(\frac{k_2}{k_2-ik_1}\right)+B = C e^{ik_1 a} e^{k_2 a}$$

$$A\left(\frac{k_2-ik_1}{k_2}\right)\left(\frac{k_2}{(k_2+ik_1)}\right)+B = C e^{ik_1 a} e^{-k_2 a}$$

$$A\left(\frac{k_2+ik_1}{(k_2-ik_1)} - \frac{(k_2-ik_1)}{(k_2+ik_1)}\right) = C e^{ik_1 a}\left(e^{k_2 a} - e^{-k_2 a}\right)$$

$$A\ \frac{k_2^2-k_1^2+2ik_1k_2-(k_2^2-k_1^2-2ik_1k_2)}{(k_2-ik_1)(k_2+ik_1)} = \qquad ||$$

$$A\ \frac{4ik_1k_2}{k_2^2+k_1^2} = C e^{ik_1 a}\left(e^{k_2 a} - e^{-k_2 a}\right)$$

$$A\ \frac{4i\frac{\cancel{2m}\sqrt{E(V_0-E)}}{\cancel{\hbar^2}}}{\frac{\cancel{2m}(V_0-E)}{\cancel{\hbar^2}}+\frac{\cancel{2m}E}{\cancel{\hbar^2}}} = \qquad ||$$

$$|A|^2\ \frac{16\,E(V_0-E)}{V_0} = |C|^2\left(e^{k_2 a} - e^{-k_2 a}\right)^2$$

IPOTESI $\Rightarrow$ TRASCURO

$$|A|^2\ \frac{16\,E(V_0-E)}{V_0} = |C|^2 e^{2k_2 a}$$

$$\frac{|C|^2}{|A|^2} = \frac{16\,E(V_0-E)}{V_0}\, e^{-2k_2 a}$$